AgriSense AI

Multi-Agent Precision Agriculture Platform

Autonomous Agents. Immutable Decisions. Empowered Farmers.

Hedera Hello Future Apex Hackathon 2026 | Theme 1: AI & Agents | March 2026

Joshua Okoghie & Favour Ogudu • University of Benin, Nigeria

Who We Are

Joshua Okoghie

Founder & CEO

Full-Stack Engineer
AI/ML Development
Hedera Integration

Favour Ogudu

Co-Founder & CDO

Data Science
Model Training
ML Architecture

🎓 Final-year Computer Science · University of Benin (UNIBEN), Nigeria

02 · THE PROBLEM

$10B+

lost annually by African smallholder farmers to preventable crop failures

🌿 Crop Losses

40-50%

Preventable pests & diseases destroy nearly half of every harvest

🌡 Climate Risk

No Warning

Droughts and erratic weather with zero early detection systems

🚫 Uninsured

>80%

Farmers cannot access agricultural insurance or credit

❓ Blind Decisions

No Data

Critical farming decisions made without reliable, verified data

The Core Gap: Existing solutions are reactive, not autonomous. Farmers get generic responses — no system thinks, decides, and acts on their behalf.

03 · OUR SOLUTION — THE 4 AGENTS

CropWatchAgent

Reactive + Autonomous

Pulls NASA HLS Sentinel-2 satellite imagery. Computes NDVI/NDWI. Logs every observation to its dedicated Hedera HCS topic.

AdvisoryAgent

Reactive

CNN(Model) disease detection (97.7% accuracy). Calls Gemini 2.5 Flash with farm context — NDVI, drought risk, location — for dynamic expert advice.

InsuranceOracleAgent

FULLY AUTONOMOUS

Polls satellite data autonomously. When NDVI < 20% + poor vegetation + high drought — calls smart contract. HBAR payout. Zero paperwork.

DataMarketplaceAgent

Reactive

Sends 1 ASAI token reward per inference via HTS. Mints a unique NFT per prediction with HIP-412 metadata on IPFS.

🎯 **Orchestrator: One API call coordinates all agents — CropWatch + Advisory in parallel — with every decision logged to dedicated HCS topics.**

04 · HEDERA INTEGRATION

📡 HCS — 4 Agent Topics

- CropWatchAgent → 0.0.8308940
- AdvisoryAgent → 0.0.8308941
- InsuranceOracleAgent → 0.0.8308942
- DataMarketplaceAgent → 0.0.8308943
- ECDSA-signed · ABFT finality · fully auditable

🪙 HTS — Token & NFT

- ASAI Utility Token: 0.0.8308875
- 1 ASAI reward per inference
- NFT collection: 0.0.8308897
- HIP-412 metadata on IPFS via Pinata
- Unique NFT minted per prediction

📜 HSCS — Smart Contracts

- PestReward: 0.0.8323760
- CropInsurance: 0.0.8323764
- ASAI Token: 0.0.8323752
- Solidity via Hardhat on Hedera testnet
- Auto HBAR payouts on drought confirmation

🔍 *Every autonomous decision is publicly verifiable on hashscan.io*

05 · TECHNICAL ARCHITECTURE

AI & ML	Custom CNN(Model) · EfficientNetB3 · 10.9M params · 50K+ PlantVillage images · 97.7% accuracy Gemini 2.5 Flash · RandomForest yield & irrigation · NASA HLS satellite NDVI/NDWI
Backend	FastAPI (Python) · Async agent loops (asyncio) · Azure Blob Storage for ML models MQTT (HiveMQ) for IoT sensors · WebSocket for real-time drone analysis
Blockchain	Hedera SDK (Node.js) · HCS + HTS + HSCS · Solidity via Hardhat IPFS via Pinata (HIP-412) · EVM address resolution via Mirror Node API
Frontend	React + TypeScript · Tailwind CSS glassmorphism · Leaflet GPS boundary drawing Real-time agent activity feed · HashScan deep links per agent topic
DevOps	AWS EC2 (t3.medium, Ubuntu 22.04) · Docker containers · GitHub Actions CI/CD Vercel for frontend · Both services live and deployed

06 · JUDGING CRITERIA ALIGNMENT

Innovation 10%

First multi-agent precision agriculture system with autonomous insurance on Hedera. Per-agent HCS topics never seen in the ecosystem.

Feasibility 10%

Fully deployed: EC2 + Docker + Vercel. All 3 smart contracts live. Real NASA satellite data.

Execution 20%

Complete working system with all endpoints, CI/CD, clean modular architecture (agents/, tools/, api/routes/, hashgraph/services/).

Integration 15%

All 3 Hedera services used deeply: HCS (4 agent topics, signed), HTS (ASAI + NFT), HSCS (3 contracts). Novel per-agent topic architecture.

Success 20%

Every prediction creates Hedera account interactions. ASAI rewards drive monthly active accounts. Insurance oracle drives contract TPS.

Validation 15%

150+ beta farmers. Live deployment during Apex period. Farmers see HCS proof of every agent decision that builds verifiable trust.

07 · DEMO & LIVE SYSTEM

🌐 **Live Demo**

https://agrisense-agents.vercel.app

⚡ **Backend API Docs**

http://44.204.70.141:8000/docs

💻 **GitHub Repo**

github.com/Bigdreams415/agrisense-agents

What the Demo Shows

🔬 Disease Detection

Upload leaf photo → CNN(Model) detects → Gemini Advisory generates contextual advice → HCS logged → ASAI reward → NFT minted

🌍 Farm Intelligence

Draw farm boundary → Orchestrator triggers CropWatch (satellite) + Advisory (disease) → unified report with 2 agent HCS proofs

🛡️ Insurance Oracle

Start autonomous loop → oracle polls satellite → drought conditions met → smart contract fires → HBAR payout triggered on-chain

🔗 On-Chain Audit

All 4 agent topic links on HashScan. Verify every autonomous decision, timestamp, and payout transaction

08 · FUTURE ROADMAP

Key Learnings

1. Per-agent HCS topics create a genuinely novel audit trail for agricultural DePIN

2. Satellite NDVI analysis works but needs edge-caching for rural 2G/3G

3. Farmers trust agent-framed advice more than generic ML output

4. Gemini context injection (NDVI + drought + location) dramatically improves advice relevance

Phase 2

Next 6 Months

- Auto-start oracle on insurance purchase
- Multi-farm portfolio monitoring
- Flutter mobile app (offline-first)
- Mainnet with real HBAR premiums

Phase 3

12 Months

- Agent marketplace for specialized agents
- 10,000 farmers: Nigeria, Ghana, Kenya
- On-chain credit scoring via farm data

Vision: Every smallholder farmer in Africa has autonomous AI agents watching their farm 24/7 — all verifiable, all transparent, all on Hedera.

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Built on Hedera. Built for Africa.

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